

# Algebra MATH-310

## Lecture 2

Anna Lachowska

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# Plan of the course

- 1 Integers: 1 lecture
- 2 Groups: 6 lectures
- 3 Rings and fields: 5 lectures
- 4 Review: 1 lecture

## Today: Groups-1

- (a) Definition and first examples
- (b) Subgroups
- (c) Cosets and Lagrange's theorem
- (d) Application: Euler's and Fermat's theorems
- (e) Application: RSA

# Groups: definition

## Definition

A **group** is a set  $G$  with a binary operation  $\cdot : G \times G \rightarrow G$  satisfying the axioms:

- 1 Associativity:  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$  for any  $a, b, c \in G$ .
- 2 Neutral element:  $\exists 1 \in G$  such that  $1 \cdot a = a \cdot 1 = a$  for any  $a \in G$
- 3 Inverse: For any  $a \in G \exists a^{-1} \in G$  such that  $a \cdot a^{-1} = a^{-1}a = 1$ .

## Definition

A group  $G$  is called **finite** if  $|G| < \infty$ . In this case  $|G| \in \mathbb{N}$  is called **the order of the group**.

## Definition

A group  $G$  is called **abelian** if  $a \cdot b = b \cdot a$  for any  $a, b \in G$ .

# Groups: first examples

- ① The real numbers  $(\mathbb{R}, +, 0)$  form an abelian group with respect to addition. The integers  $(\mathbb{Z}, +, 0)$  form an abelian group with respect to addition. *infinite abelian*
- ② For any  $n \in \mathbb{N}, n \geq 2$ , the equivalence classes of integers modulo  $n$ :  $\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \dots, [n-1]\}$   *$n=6$*  form an abelian group with respect to addition. In  $\mathbb{Z}/6\mathbb{Z}$ , we have *finite abelian*  $[2] + [5] = [1]$  etc. The order  $|\mathbb{Z}/n\mathbb{Z}| = n$ .  
 *$[0]$  is neutral ;  $[2] + [4] = [0] \Rightarrow [4]$  is the inverse of  $[2]$  in  $\mathbb{Z}/6\mathbb{Z}$*
- ③ The group of real invertible  $n \times n$  matrices  $GL(n, \mathbb{R})$  is a non-abelian infinite group with respect to the matrix multiplication.

$$A \cdot B \neq B \cdot A; \quad \forall A \exists A^{-1} \in GL(n, \mathbb{R}). \quad A \cdot A^{-1} = A^{-1} \cdot A = \text{Id}$$

$$\text{Id} = \begin{pmatrix} 1 & & & 0 \\ & 1 & & \\ & & 1 & \\ 0 & & & 1 \end{pmatrix}$$

# Groups: further examples

## Multiplicative group modulo $n$

Let  $n \in \mathbb{Z}_{\geq 2}$ ,  $K = \{x \in \mathbb{N} : 1 \leq x \leq n, \text{gcd}(x, n) = 1\}$ . Then  $K$  is a group with respect to multiplication modulo  $n$ , and  $|K| = \varphi(n)$ . (by def)

Notation:  $(\mathbb{Z}/n\mathbb{Z}, \cdot)^* = ((\mathbb{Z}/n\mathbb{Z})^*, \cdot)$

$$\begin{aligned} \{1 \leq x \leq n : \text{gcd}(x, n) = 1\} &\Leftrightarrow \exists a, b \in \mathbb{Z} : ax + bn = 1 \Leftrightarrow \\ ax &\equiv 1 \pmod{n} \Leftrightarrow [a] \cdot [x] = [1] \\ \Rightarrow [x] &\text{ is invertible in } (\mathbb{Z}/n\mathbb{Z}, \cdot)^* \end{aligned}$$

Let  $n=5$  Then  $K = \{[1], [2], [3], [4]\}$

$$[1] \cdot [1] = [1] \quad ; \quad [2] \cdot [3] = [1] \quad , \quad [4] \cdot [4] = [1] \Rightarrow$$

$$[2]^{-1} = [3] \quad , \quad [3]^{-1} = [2] \quad , \quad [4]^{-1} = [4]$$

$[1]$  is the neutral element in  $K$ .

# Conclusions

- 1 There are groups with respect to addition, multiplication, or another binary operation satisfying the axioms.
- 2 Important example: additive and multiplicative groups of integers modulo  $n$ :  $(\mathbb{Z}/n\mathbb{Z}, +) \neq (\mathbb{Z}/n\mathbb{Z}, \cdot)^*$ .  
In particular,  $|(\mathbb{Z}/n\mathbb{Z}, +)| = n$  and  $|(\mathbb{Z}/n\mathbb{Z}, \cdot)^*| = \varphi(n)$ .



see Appendix B  
groups - Math 310.pdf  
for more examples  
of unusual groups  
(Appendices A and B  
are not part of the exam)

# Subgroups

## Definition

A **subgroup**  $H \subset G$  is a subset in  $G$  such that  $1 \in H$  and  $H$  is closed with respect to the multiplication and taking inverses.

Example:  $\{0, \pm 3, \pm 6, \pm 9, \dots\} \subset (\mathbb{Z}, +, 0)$  is a subgroup of integers with respect to addition.

$$3n + 3k = 3(n+k) \in \{0, \pm 3, \pm 6, \dots\}$$

$$3n + 3(-n) = 0 \quad \text{inverse element}$$

$$0 \quad \text{neutral element}$$

## Subgroup generated by a single element

Suppose  $g \in G$ . Consider the subset  $\langle g \rangle = \{1, g^{\pm 1}, g^{\pm 2}, g^{\pm 3}, \dots\} \subset G$ . Then  $g^i \cdot g^k = g^{i+k} \in \langle g \rangle$ , and for each  $g^i$  the inverse  $g^{-i} \in \langle g \rangle$ . Therefore  $\langle g \rangle \subset G$  is a subgroup by construction. It is called the **subgroup in  $G$  generated by the element  $g$** . It is the smallest subgroup of  $G$  containing  $g$ .

Example:  $(\mathbb{Z}, +, 0) = G$ ,  $g = 4 \Rightarrow \langle g \rangle = \{0, \pm 4, \pm 8, \pm 12, \dots\} = 4\mathbb{Z}$   
 $4\mathbb{Z} \subset \mathbb{Z}$  is a subgroup with respect to addition generated by  $4 \in \mathbb{Z}$ .

### Definition

If there exists  $n \in \mathbb{N}_+$  minimal such that  $g^n = 1$  in  $G$ , then  $n$  is called **the order of element  $g$** . In this case  $\langle g \rangle = \{1, g, g^2, \dots, g^{n-1}\}$  and  $|\langle g \rangle| = n$ .

# Cosets

## Definition

Let  $H \subset G$  be a subgroup and  $g \in G$  an element. The **left coset**  $gH$  is the set of group elements of the form  $gH = \{gh, h \in H\}$ .

## Proposition

- 1 Two cosets  $xH$  and  $yH$  are either equal or disjoint:  $xH = yH$  or  $xH \cap yH = \emptyset$ .
- 2 Any element  $g \in G$  belongs to a left  $H$ -coset
- 3 If  $H$  is finite, then  $|xH| = |H|$  for any  $x \in G$ .

Proof: (1) If  $xH \cap yH \neq \emptyset \Rightarrow \exists h_1, h_2 \in H : xh_1 = yh_2 \Rightarrow x = yh_2h_1^{-1} = yh_3 \in yH$   
Then  $\forall h \in H \Rightarrow xh = y\underbrace{h_3h}_\in H \in yH \Rightarrow xH \subset yH$  similarly  $yH \subset xH \Rightarrow xH = yH$ .

(2) Take the coset of  $g$ :  $gH = \{g, gh_1, gh_2, \dots\}$

(3) Let  $f: H \rightarrow xH$ ,  $f(h) = xh \Rightarrow f$  is surjective:  $xH = \{xh\}_{h \in H}$   
 $\forall h \in H$   $f$  is injective: if  $xh_1 = xh_2 \Rightarrow x^{-1}xh_1 = x^{-1}xh_2 \Rightarrow h_1 = h_2$

**Example of cosets**  $(\mathbb{Z}, +, 0) = G$  ;  $H = 4\mathbb{Z} \subset \mathbb{Z}$  subgroup.

Coset of 0 wrt  $4\mathbb{Z}$  in  $\mathbb{Z}$ :

$$\{0+4k\}_{k \in \mathbb{Z}} = 4\mathbb{Z} = H = \{0, \pm 4, \pm 8, \dots\}$$

Coset of 1 wrt  $4\mathbb{Z}$  in  $\mathbb{Z}$ :

$$\{1+4k\}_{k \in \mathbb{Z}} = \{1, 5, 9, -3, -7, \dots\} \text{ left coset in } \mathbb{Z}$$

Coset of 5 wrt  $4\mathbb{Z}$  in  $\mathbb{Z}$ :

$$\{5+4k\}_{k \in \mathbb{Z}} = \{1, 5, 9, -3, -7, \dots\} = \{1+4k\}_{k \in \mathbb{Z}} \text{ same coset}$$

Coset of 2 wrt  $4\mathbb{Z}$  in  $\mathbb{Z}$ :

$$\{2+4k\}_{k \in \mathbb{Z}} = \{\pm 2, \pm 6, \pm 10, \dots\} \text{ left coset in } \mathbb{Z}$$

Note that  $\mathbb{Z} = \{0+4k\}_{k \in \mathbb{Z}} \cup \{1+4k\}_{k \in \mathbb{Z}} \cup \{2+4k\}_{k \in \mathbb{Z}} \cup \{3+4k\}_{k \in \mathbb{Z}}$

# Lagrange's theorem

## Theorem

Let  $G$  be a finite group and  $H \subset G$  a subgroup. Then  $|H|$  divides  $|G|$ .

Proof: Each  $g \in G$  belongs to a left  $H$ -coset, and  $\begin{cases} xH = yH, \text{ or} \\ xH \cap yH = \emptyset \end{cases}$

$\Rightarrow G = \bigcup_{i=1}^r x_i H$  disjoint union of finitely many left  $H$ -cosets

$$|G| = \sum_{i=1}^r |x_i H| \quad ; \quad |x_i H| = |H| \quad \forall x_i$$

$$\Rightarrow |G| = \sum_{i=1}^r |H| = r \cdot |H| \Rightarrow |H| \text{ divides } |G|$$



## Definition

The number of left  $H$ -cosets in  $G$  is called **the index of  $H$  in  $G$** . Notation:

$$[G : H] = \frac{|G|}{|H|} \in \mathbb{N}_+.$$

# Order of an element divides order of the group

## Corollary

- 1 Let  $G$  be a finite group, and  $g \in G$ . Then the order of  $g$  divides  $|G|$ .
- 2  $g^{|G|} = 1$ .

Proof: (1) Let  $H = \langle g \rangle = \{1, g, g^2, \dots, g^{k-1}\}$  where  $k$  is the order of the element  $g$  in  $G$ .  
 $\Rightarrow \langle g \rangle = H \subset G$  subgroup  $\Rightarrow$  by Lagrange  $\Rightarrow |\langle g \rangle| = k$  divides  $|G|$ .

(2) We have  $g^k = 1$ , also  $|G| = kt$  where  $t \in \mathbb{N}$

$$\Rightarrow g^{|G|} = g^{kt} = (g^k)^t = 1^t = 1 \in G$$



# Applications of Lagrange's theorem

## Theorem

### (Euler's theorem)

Let  $a, n \in \mathbb{Z}_+$  such that  $\gcd(a, n) = 1$ . Then  $a^{\varphi(n)} \equiv 1 \pmod{n}$ .

Proof: Let  $G = (\mathbb{Z}_n^\times, \cdot, 1)$ . Then  $|G| = \varphi(n)$   
if  $\gcd(a, n) = 1 \Rightarrow [a] \in G \Rightarrow$  by Corollary  $[a]^{\varphi(n)} = [1] \in G$   
 $\Leftrightarrow a^{\varphi(n)} \equiv 1 \pmod{n}$



## Theorem

### (Fermat's little theorem)

Let  $a \in \mathbb{Z}_+$  and  $p$  a prime that does not divide  $a$ . Then  $a^{p-1} \equiv 1 \pmod{p}$ .

Proof:  $\gcd(a, p) = 1$ ,  $\varphi(p) = p-1 \Rightarrow$  By Euler's thm  $\Rightarrow a^{\varphi(p)} = a^{p-1} \equiv 1 \pmod{p}$

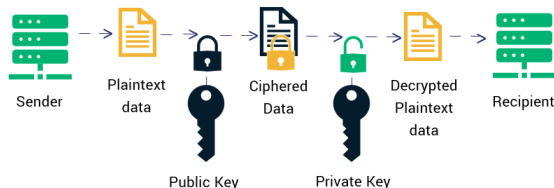
# Conclusions

- ① If  $G$  is a finite group, and  $H \subset G$  a subgroup, then  $|H|$  divides  $|G|$ .
- ② If  $G$  is a finite group, and  $g \in G$ , then the order of the element  $g$  divides  $|G|$ .
- ③ Let  $a, n \in \mathbb{Z}_+$  such that  $\gcd(a, n) = 1$ . Then  $a^{\varphi(n)} \equiv 1 \pmod{n}$ .

Why do we care?

*Rivest-Shamir-Adleman 1977*

## How RSA Encryption Works



# RSA encryption system

## Setting

- 1 Choose two large distinct primes  $p, q$ .  $\sim 2^{1030}$
- 2 Let  $m = pq$ . Then  $\varphi(m) = (p-1)(q-1)$ .  
*factorization is HARD*
- 3 Choose  $1 < e < \varphi(m)$  such that  $\gcd(e, \varphi(m)) = 1$ .
- 4 Use the Euclidean algorithm to find  $d \in \mathbb{Z}$  such that  $ed - k\varphi(m) = 1$  for some  $k \in \mathbb{Z}$ .
- 5 Publish the encryption key  $(m, e)$ .
- 6 Keep secret the decryption key  $(m, d)$ .

## Send a message

- 1 **A** publishes the encryption key  $(m, e)$ .
- 2 **B** wants to send message  $x$ ,  $0 < x < m$  to **A**. Then **B** computes  $y \equiv x^e \pmod{m}$  and sends  $y$  publicly to **A**.  
*HARD*
- 3 **A** computes  $y^d = x^{ed} \equiv x \pmod{m}$ .

# RSA encryption system

Why does it work?

## Proposition

Let  $p, q$  be two distinct primes and  $m = pq$ . Let  $1 < e < \varphi(m)$  be such that  $\gcd(e, \varphi(m)) = 1$ , and  $d \in \mathbb{Z}$  such that  $ed - k\varphi(m) = 1$  for some  $k \in \mathbb{Z}$ . Then for any  $0 < x < m$ ,  $x^{ed} \equiv x \pmod{m}$ .

Proof: (1) If  $x = pt \Rightarrow x^{ed} \equiv 0 \pmod{p} \Rightarrow (x^{ed} - x) \equiv 0 \pmod{p}$ .

(2) If  $x$  is not divisible by  $p$ .  $\Rightarrow$  Fermat's thm  $\Rightarrow x^{p-1} \equiv 1 \pmod{p}$

$$x^{ed} = x^{k\varphi(m)+1} = x^{k(p-1)(q-1)+1} = \underbrace{(x^{p-1})^{k(q-1)}}_{\equiv 1 \pmod{p}} \cdot x \equiv 1 \cdot x \pmod{p}$$

$$\Rightarrow (x^{ed} - x) \equiv 0 \pmod{p}$$

The same argument works for  $q \Rightarrow (x^{ed} - x)$  divisible by  $p$  and  $q$

$$\Rightarrow x^{ed} \equiv x \pmod{\underset{m}{pq}}$$



## RSA encryption system: example

**Setting:** Let  $p, q$  be two distinct primes and  $m = pq$ . Let  $1 < e < \varphi(m)$  be such that  $\gcd(e, \varphi(m)) = 1$ , and  $d \in \mathbb{Z}$  such that  $ed - k\varphi(m) = 1$  for some  $k \in \mathbb{Z}$ . Then for any  $0 < x < m$ ,  $x^{ed} \equiv x \pmod{m}$ .

Example  $p=5, q=7 \Rightarrow m=pq=35, \varphi(m)=(p-1)(q-1)=24$

Let  $e=5 \Rightarrow \gcd(5, 24)=1$  Find  $d$ :  $ed - k\varphi(m) = 1$

$$5 \cdot 5 - 24 \cdot 1 = 1 \Rightarrow d=5 \quad \begin{array}{l} (m, e) = (35, 5) \text{ encoding key} \\ (m, d) = (35, 5) \text{ decoding key.} \end{array}$$

Send  $x=31$ . Compute  $x^e \pmod{m}$

$$\begin{aligned} 31^5 \pmod{35} &= (-4)^5 \pmod{35} = -64 \cdot 16 \pmod{35} = 6 \cdot 16 \pmod{35} \\ &= 32 \cdot 3 \pmod{35} = -3 \cdot 3 \pmod{35} = -9 \pmod{35} \end{aligned}$$

$\rightarrow$  Send  $y = -9$ . or  $y = 26$

## RSA encryption system: example

**Setting:** Let  $p, q$  be two distinct primes and  $m = pq$ . Let  $1 < e < \varphi(m)$  be such that  $\gcd(e, \varphi(m)) = 1$ , and  $d \in \mathbb{Z}$  such that  $ed - k\varphi(m) = 1$  for some  $k \in \mathbb{Z}$ . Then for any  $0 < x < m$ ,  $x^{ed} \equiv x \pmod{m}$ .

To decode:  $y^d \pmod{m}$   $(m, d) = (35, 5)$

$$(-9)^5 \pmod{35} = 81 \cdot 81 \cdot (-9) \pmod{35} = -11 \cdot 11 \cdot 9 \pmod{35}$$

$$= -99 \cdot 11 \pmod{35} = -29 \cdot 11 \pmod{35} = 6 \cdot 11 \pmod{35} =$$

$$= 66 \pmod{35} \equiv 31 \pmod{35}$$

$$\Rightarrow x = 31. \quad \text{😊}$$